# Self-Supervised Learning of Contrast-Diffusion Models for Land Cover Classification in SAR Images

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***Abstract*—Deep learning (DL) methods have been widely applied to synthetic aperture radar (SAR) land cover classification. The complexity of SAR data and the limited availability of labeled samples greatly constrain the feature learning and the generalization ability of the model. Inspired by the excellent generative performance of denoising diffusion probabilistic models (DDPMs) on complex data distributions, a self-supervised learning framework based on contrast-diffusion models (CDMs) is proposed to expand the applicability to multiple broad scenarios with complex and varying imaging conditions under limited annotated data conditions. Specially, the proposed framework consists of the upstream CDM pretraining on all unlabeled samples and the downstream land cover classification with few labeled samples in each test scene. Concretely, in the upstream task, the features are captured through the generative learning of the DDPM. Following this, the dimensionality reduction and resolution expansion (DRRE) module is designed and embedded to reduce feature redundancy and align the feature granularity between layers and the input image. Finally, contrastive learning is employed to enforce semantic feature consistency across different steps. In the downstream task, the feature in pretrained CDM is efficiently delivered in a single-step reverse diffusion process and then fine-tuned with few labeled samples from each test scene and finally output the predictions. Compared with several supervised and self-supervised methods, the proposed framework achieves superior classification and generalization performance on multiple broad scenes with complex and varying imaging conditions. For example, based on the average results from six test scenes, CDM shows improvements in overall accuracy (OA) of 41.42%, 34.40%, 7.70%, 8.53%, and 9.60% compared to DeepLabv3+, CCNR, SegFormer, MAE, and DDPM, respectively. The code is available at https://github.com/gosling123456/CDM.git**

***Index Terms*—Contrast learning, diffusion models, land cover classification, self-supervised learning (SSL), synthetic aperture radar (SAR).**

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## I. Introduction

In recent years, synthetic aperture radar (SAR) has brought a vast amount of radar images with high resolution, wide coverage, and multiple imaging modes for Earth remote sensing. These high-quality data have greatly advanced remote sensing understanding and interpretation tasks such as land cover classification, object detection and recognition, and change detection in SAR images [1]. SAR image land cover classification is a fundamental task with diverse applications in land cover analysis [2], polar monitoring [3], resource detection [4], and geospatial analytics [5].

Throughout the classification process, the discriminative capability and generality of the feature representation play a pivotal role in determining the accuracy and applicability of the model. In the past few decades, a variety of classification methods have been developed to accommodate the diverse features present in SAR terrain features. Statistical modeling of intensity information from various terrain categories in SAR images can offer valuable terrain insights, particularly for low-resolution SAR images. Based on this, Weibull [6], Fisher [7], Rician [8], and Gaussian mixture [9] distributions are used for SAR image intensity statistical feature extraction. Texture is a critical visual characteristic that varies based on systems and exhibits distinctive textural properties across different types of terrain surfaces. Gray-level co-occurrence matrix (GLCM) [10], gray-level gradient co-occurrence (GLGCM) [11], and multilevel local pattern histogram (MLPH) [12] are commonly used to capture SAR image texture features. The methods mentioned primarily concentrate on the low-level and mid-level features of SAR images, with perceptual capabilities typically confined to individual pixels or local regions. These approaches often lack the capacity to comprehensively describe overall and semantic information of the images.

Contextual features extracted from SAR images highlight the interplay of terrain characteristics, enabling breakthrough results in land cover classification when integrated with deep learning (DL) methodologies. Geng et al. [13] proposed utilizing long short-term memory (LSTM) to extract contextual dependencies from SAR images, and then combined this approach with non-negative and Fisher-constrained autoencoders (NF-CAEs) to enhance feature discrimination for classification. Zhao et al. [14] employed ensemble learning (EL) and deep belief networks (DBNs) to extract

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discriminative features from SAR images. Liang et al. [15] proposed a context-aware encoder network (CAEN) to capture local-to-global semantic contextual features of high-resolution SAR images for classification. Relying on extensive manually labeled training data, DL has emerged as a mainstream classification method in remote sensing [16], [17], [18]. However, labeling data is expensive and time-consuming, particularly for SAR images featuring large scenes, high resolution, and complex terrain characteristics and imaging processes. In addition to this, the distribution of SAR image pixel values is significantly impacted by imaging systems, leading to substantial domain discrepancies among SAR images captured from various scenes. This attribute of SAR images confines the training and testing of state-of-the-art DL algorithms for land cover classification to the same scenes or imaging systems [19]. To address these challenges, [20] and [21] are exploring two directions: domain adaptation and transfer learning. However, these approaches remain limited. Overall, the complexity of SAR data and the limited availability of labeled samples continue to pose significant challenges to prevailing land cover classification methods.

Self-supervised learning (SSL), as a minimal labeling paradigm, offers significant advantages in overcoming the scarcity of SAR image labels and improving model generalizability. SSL leverages unlabeled SAR images from different domains to extract visual features through a series of pretext tasks, enabling the application of these learned features to specific downstream tasks. Contrastive SSL [22] learns feature representations by making positive pairs similar and negative pairs dissimilar. Generative SSL acquires feature representations by generating or reconstructing either partial or entire input images, with pretext tasks such as image restoration [23], image super-resolution [24], and image colorization [25]. Zhang et al. [26] and Wang et al. [27] employ a generative adversarial network (GAN) and a masked autoencoder in their generative pretraining approach to acquire image features. In our previous research [28], a style transfer generation task was used to learn the content features of different terrains of SAR images. These efforts have achieved success in land cover classification. However, the inconsistency in learning objectives between upstream and classification tasks can lead to differing guidance for feature learning, potentially degrading classification performance.

The denoising diffusion probabilistic model (DDPM) [29] has achieved significant success in image generation by producing diverse and realistic image samples from Gaussian noise through a sequence of denoising steps. This model holds promising application potential for addressing the complexities inherent in data distribution. Perera et al. [30] employed the DDPM for SAR image denoising, illustrating its capability to faithfully capture diverse terrain features in SAR images across varying noise levels through a multistep denoising process. Baranchuk et al. [31] examined the semantic information conveyed by network intermediate activation features during the backpropagation process of diffusion models across varying steps. They demonstrated that these intermediate activations effectively represent the semantic details of input images. However, they did not provide a satisfactory explanation for the effectiveness of these intermediate activations in capturing semantic information. Yang and Wang [32] conducted an extensive investigation into the representational capabilities of the diffusion model, revealing that it is inherently a denoising autoencoder. Chen et al. [33] investigated the function of various components in SSL by deconstructing the diffusion model. They demonstrated that the noise-added low-dimensional potential space plays a crucial role in enhancing competitiveness in SSL.

Inspired by the above studies, this article proposes a contrast-diffusion model (CDM) framework for large-scale generative self-supervised learning pretraining and land cover classification of SAR images. The CDM learns feature representations of terrain from SAR images by modeling SAR data distributions using large-scale unlabeled datasets, reducing the dependence on labeled data and enhancing feature targeting and generalization capabilities. To be specific, the method employs a multistep denoising process to progressively decouple speckle noise from land cover information, enabling the learning of pure and distributed feature representations across varying noise levels. Furthermore, the module of DRRE is constructed to realize feature dimensionality reduction and splicing, maintaining the resolution for the land cover classification in a parameter-free manner. Finally, contrastive learning is integrated to align shared spatial features of land covers across diffusion steps, enhancing their discrimination for classification tasks. The resulting CDM model is pretrained on large-scale and unlabeled SAR datasets with diverse imaging parameters, capturing both intrinsic features of the same land cover type and discriminative features between different types. As a result, the model delivers superior classification performance while reducing reliance on labeled data.

Three main contributions are summarized as follows.

1) A mixed self-supervised learning framework using both generative and discriminative modes is proposed to capture intrinsic SAR image terrain feature with a minimal amount of labeled samples.
2) Integrating contrast learning into the generation process enhances the consistency of terrain semantic features across varying noise levels and naturally bridges the gap between the directionless pretraining and the intended classification tasks.
3) Compared with several supervised and self-supervised methods, the proposed framework achieves superior classification and generalization performance on multiple broad scenes with complex and varying imaging conditions.

The remainder of this article is organized as follows. Section II reviews the DDPM. The implementation details of the proposed CDM are unfolded in Section III, which includes the complete process of pretraining and land cover classification. Section IV reports the results of the SAR data generation and classification experiments and validates the effectiveness of the proposed method. Section V concludes the article.

## II. Recap on DDPM

The DDPM is rooted in nonequilibrium thermodynamics and functions as a probabilistic generative model. The core

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concept involves initiating with random noise and progressively eliminating and refining this noise across several defined steps until a clean generated image is produced. This ensures that any small errors that occur early on can be corrected in subsequent steps. Specifically, the diffusion model learns the distribution $p_{x_0}$ of the training image $x_0$ through variational inference on a Markov process with $T$ steps. This process is divided into two main parts: the forward diffusion process and the reverse denoising process.

### A. Forward Diffusion Process

In the forward diffusion process, an isotropic Gaussian distribution is generated by adding Gaussian noise with zero mean and variance $\beta_t$ to $x_0$ for $T$ diffusion steps until the image is completely corrupted. The forward process at each step $t$ is defined as follows:

$$q(x_t \mid x_{t-1}) = \mathcal{N}\left(x_t;\ \sqrt{1-\beta_t}x_{t-1}, \beta_t \mathbf{I}\right) \tag{1}$$

where $(x_0, x_1, \ldots, x_T)$ represents a Markov chain of $T$ steps, and $(\beta_1, \ldots, \beta_T)$ is the variance schedule that controls the noise added at each step. Equation (1) can be simplified using reparameterization, enabling the acquisition of an arbitrarily noisy image $x_t$ through a single forward diffusion process. It is further expressed as follows:

$$q(x_t \mid x_0) = \mathcal{N}\left(x_t;\ \sqrt{\bar{\alpha}_t}x_0, (1-\bar{\alpha}_t)\mathbf{I}\right) \tag{2}$$

where $\bar{\alpha}_t = \prod_{i=1}^{T} \alpha_i$ and $\alpha_i = 1 - \beta_i$. $x_t$ can be further parameterized as follows:

$$x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1-\bar{\alpha}_t}\epsilon, : \epsilon \sim \mathcal{N}(0,1). \tag{3}$$

### B. Reverse Denoising Process

In the reverse denoising process, a neural network $f_\theta$ is employed to execute a sequence of minor denoising operations to derive $x_0$. At each step of the reverse process, $x_t$ undergoes a denoising operation to yield $x_{t-1}$. Thus, $f_\theta$ is utilized to mathematically represent the reverse distribution $p(x_{t-1} \mid x_t)$, defined as follows:

$$p(x_{t-1} \mid x_t) = \mathcal{N}\left(x_{t-1}; \mu_\theta(x_t, t), \sum\nolimits_\theta (x_t, t)\right) \tag{4}$$

where $\mu_\theta(x_t, t)$ and $\sum_\theta(x_t, t)$ represent the mean and variance of the $x_{t-1}$ output from $f_\theta$ given inputs $x_t$ and $t$, respectively.

In the DDPM, $f_\theta$ does not forecast the mean of $p(x_{t-1} \mid x_t)$ as in (4), but instead the $\mu$ of the noise component at step $t$. This distinction arises because the mean of $x_{t-1}$ can be derived through a linear combination of $\mu$ and $x_t$.

Building upon the forward and reverse processes described above, the parameters of $f_\theta$ are determined by minimizing the KL divergence between the forward and reverse distributions across all time steps. The DDPM simplifies the training objective by proposing to optimize $f_\theta$ through the minimization of the reweighted evidence lower bound of the fitted noise, defined as follows:

$$\mathcal{L}_{\text{simple}} = \mathbb{E}_{t,x_0,\epsilon}\left[\|\epsilon - f_\theta(x_t, t)\|^2\right] \tag{5}$$

where $\epsilon$ is the random noise sampled from $\mathcal{N}(0,1)$.

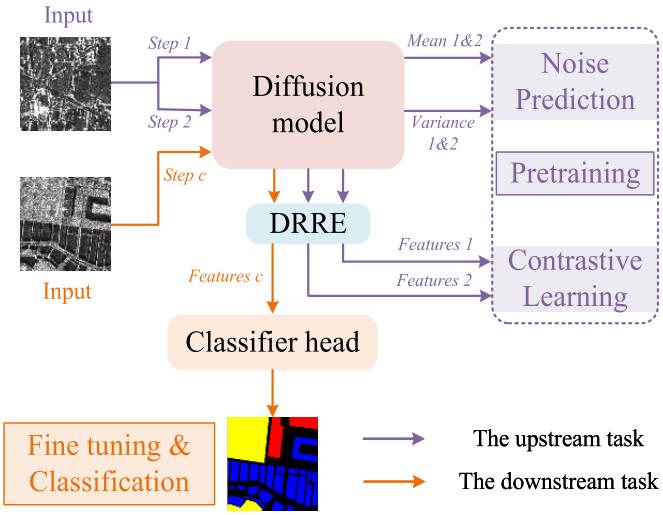



Fig. 1. Training of the CDM framework involves two tasks: the upstream and the downstream. During the upstream task, the diffusion model predicts the noisy information of the inputs and provides feature activation of input pairs for contrastive learning. In the downstream task, the learning and land cover classification process is completed.

In the inference phase, the DDPM conducts a sequence of reverse denoising processes to generate clear images from Gaussian noise. Specifically, the inference begins with Gaussian noise $x_T$, and at each step $t$, $x_t$ progressively removes the noise and yields $x_{t-1}$ according to the following equation:

$$x_{t-1} = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} f_\theta(x_t, t)\right) + \sigma_t z \tag{6}$$

where $z$ is a random variable sampled from the standard normal distribution; $t = T, \ldots, 1$, $z \sim \mathcal{N}(0,1)$, and $\sigma_t$ is the standard deviation of the reverse distribution.

## III. CDM Framework

In this section, we provide a detailed description of the proposed CDM framework, encompassing both the pretraining and land cover classification components, as shown in Fig 1. First, a critical submodule known as dimensionality reduction and resolution expansion (DRRE) is introduced. Then, this section elaborates on the implementation of a contrast learning strategy based on DRRE for pretraining the CDM. Finally, the pretrained CDM is transferred to a deep land cover classification network, and fine-tuning is performed using limited labeled data.

### A. DRRE

Deep semantic segmentation networks like U-Net [34] consist of multiple layers with downsampling and upsampling modules. The downsampling module expands the model's receptive field, allowing it to capture feature information across different scales within the image. The downsampling module efficiently captures high-level semantic features from an image but consistently decreases the resolution of these features. In semantic segmentation tasks, it is essential to restore the full resolution to accurately classify each pixel. Under supervised learning, feature resolution can be learned

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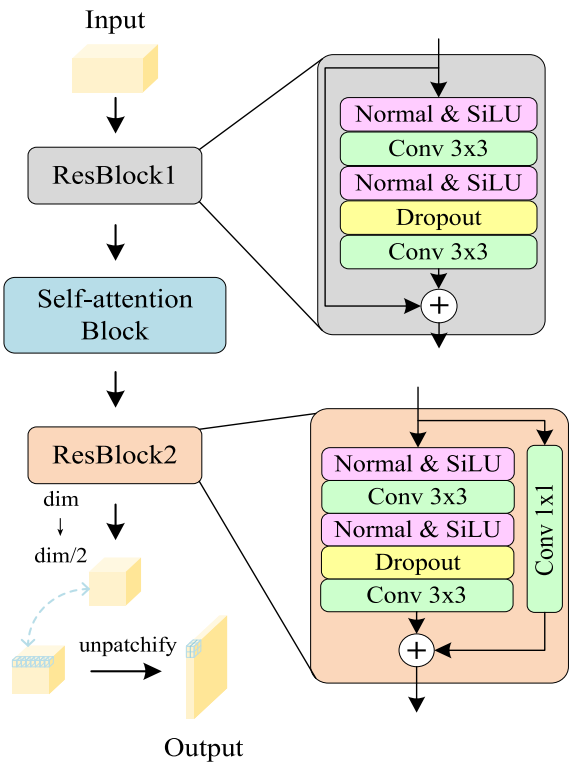



Fig. 2. Structure of DRRE.

and restored using transposed convolution [35] layers. Nevertheless, this approach can also lead to an increase in network parameters, potentially exacerbating the risk of overfitting when labeled data is insufficient. In this case, parameter-free upsampling methods like inverse pooling and interpolation offer a preferable approach. Due to various factors like the imaging system and terrain relief, SAR images frequently exhibit substantial speckle noise. The above-described parameter-free upsampling methods can recover feature resolutions, but they may not be ideal for the unique structural and noise characteristics of SAR images. Therefore, we design a new module named DRRE to address this problem.

As shown in Fig. 2, the DRRE consists of two ResBlocks and a self-attention [36] block. The self-attention block comprises several parallel attention heads. Each head computes an independent attention distribution that highlights different subspace features of the input sequence through distinct linear transformations. The outputs from each head are concatenated and integrated to create a multiperspective feature of the self-attention distribution, enhancing the model's representation and its ability to capture complex dependencies. These two residual blocks serve to deepen the network, with the second block applying dimensionality reduction to the input features, and the feature map $F_{\text{in}} \in R^{C\times h\times w}$ is transformed into $F_{\text{mid}} \in R^{(C/2)\times h\times w}$. This reduces redundant information in features and effectively reduces computational overhead in downstream land cover classification tasks. Drawing inspiration from the concept of vector-to-image splicing by MAE [37], DRRE extends feature resolution by splicing dimensionally reduced features on a channel-by-channel basis, achieving this in a parameter-free manner. The output feature map changes from $F_{\text{mid}} \in R^{(C/2)\times h\times w}$ to $F_{\text{out}} \in R^{C'\times H\times W}$, where $H$ and $W$ are the height and width of the input SAR image, $h$ and $w$ are the width and height of the input feature map, respectively, and $C' = (Chw/HW)$.

DRRE conducts feature dimensionality reduction, and avoids unsuitable upsampling methods like interpolation or pooling, common in SAR image processing, by enhancing feature resolution through channel-by-channel splicing.

### B. Pretrained CDM

The pretrained CDM is constructed based on the DDPM training process, and its single-step reverse denoising process in the training phase is shown in Fig. 3.

During pretraining, the forward diffusion process of SAR images exists for two randomly and uniformly sampled steps $t_1$ and $t_2$ from the total step $T$. Noisy images $x_{t_1}$ and $x_{t_2}$ are then generated under the same Gaussian noise as described in 3. In the reverse denoising process, $x_{t_1}$ and $x_{t_2}$ are simultaneously fed into the parameter-sharing U-Net structure of the noise prediction network $f_\theta$ and the noise is predicted. When $t_1 = t_2$, the model compares the same image at two distinct time steps, leading to identical feature activations for the two noisy images. The feature similarity loss $\mathcal{L}_{\text{sim}}$ becomes zero. However, the occurrence and influence is negligible. Furthermore, IDDPM [38] demonstrated that employing learnable noise covariance and cos noise scheduling, instead of fixed covariance and linear noise scheduling as in DDPM, enhances model performance in generating images. Therefore, in the pretraining of CDM, cos noise scheduling is employed to regulate the noise level added during the diffusion process, defined as follows:

$$\bar{\alpha}_t = \frac{f(t)}{f(0)}, f(t) = \cos\left(\frac{t/T + s}{1+s}\cdot\frac{\pi}{2}\right)^2 \tag{7}$$

where $s = 0.008$. From 7, it is possible to further calculate $\beta_t$, where $\beta_t = 1-(\bar{\alpha}_t/\bar{\alpha}_{t-1})$. In practice, $\beta_t$ is truncated to a value no greater than 0.999 to avoid singularities as the diffusion step $t$ approaches $T$.

Following the introduction of the learnable variance, $f_\theta$ not only estimates the noise mean but also outputs a vector $v$ that encapsulates each component of the noise variance. The variance is expressed as follows:

$$\sum\nolimits_\theta(x_t, t) = \exp\left(v\log\beta_t + (1-v)\log\tilde{\beta}_t\right)$$
$$\tilde{\beta}_t = \frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t}\beta_t. \tag{8}$$

Given that $\mathcal{L}_{\text{simple}}$ in the DDPM is independent of the variance term in $f_\theta$, $f_\theta$ is optimized using the hybrid loss $\mathcal{L}_{\text{hybrid}}$, expressed as follows:

$$\mathcal{L}_{\text{hybrid}} = \mathcal{L}_{\text{simple}} + \lambda_v\mathcal{L}_{vlb} \tag{9}$$

where $\mathcal{L}_{vlb}$ is a variational lower bound loss term obtained by computing the KL divergence of two Gaussian distributions, the true $x_t$ and the predicted $x_t$, and $\lambda_v$ is the weight of $\mathcal{L}_{vlb}$ that prevents it from overpowering $\mathcal{L}_{\text{simple}}$. Simultaneously, $\mathcal{L}_{vlb}$ blocks the gradient directed back to the predicted mean, ensuring that $\mathcal{L}_{\text{simple}}$ remains the primary contributor to the predicted mean gradient.

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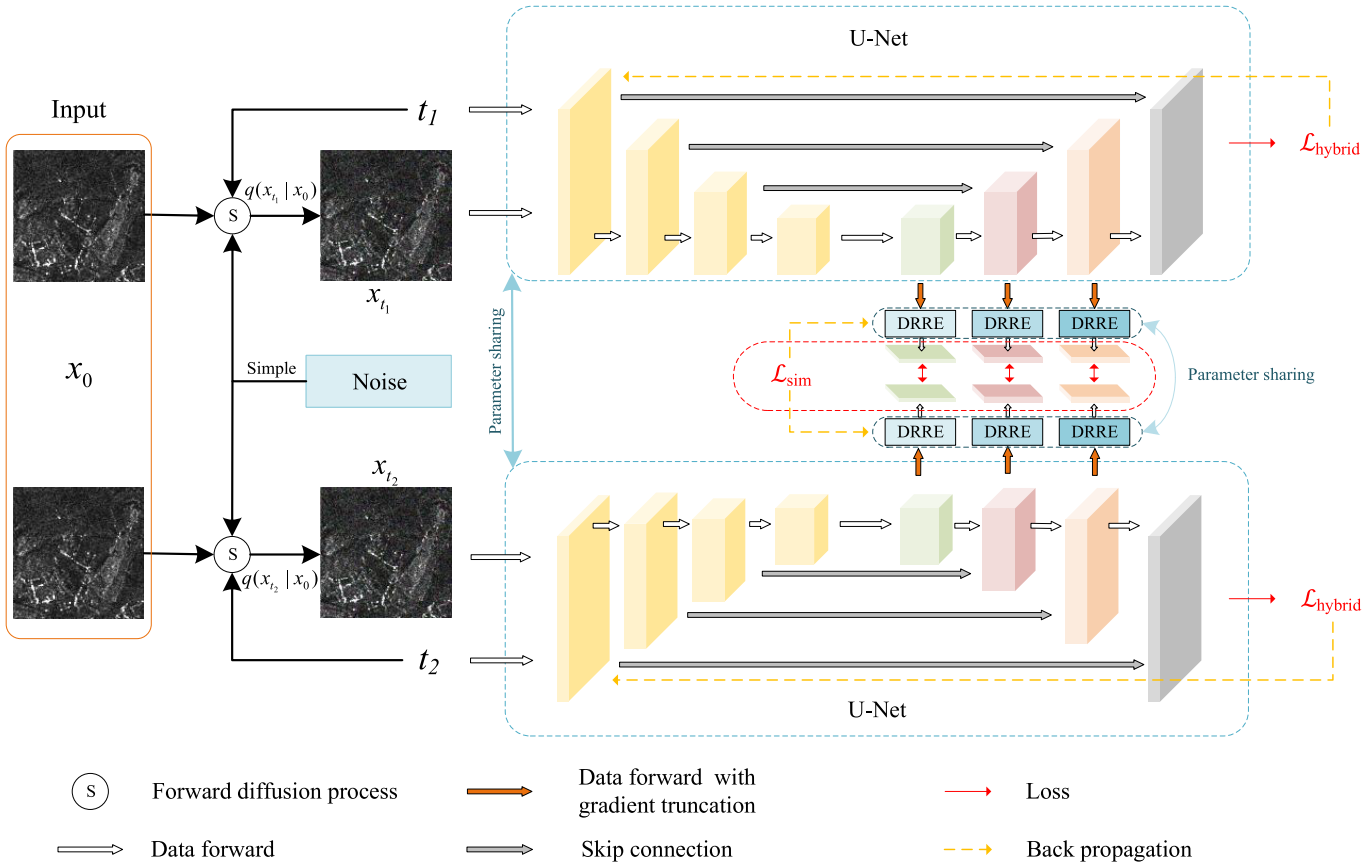



Fig. 3. Pipeline of the pretrained CDM single-stage reverse denoising process.

There exists a distinct separation between the pretraining process of the diffusion model described earlier and the downstream land cover classification task. On one hand, the varying task objectives can lead to feature representations that lack generalizability across different tasks. On the other hand, the pretraining model does not explicitly offer an interface for the downstream task. The CDM proposed in this study extends the idea of diffusion training by incorporating contrast learning. This addition aims to ensure consistent feature activation across different diffusion steps, thereby improving the coherence of terrain features in images with different noise levels and increasing the discriminative of SAR image features. Specifically, the inverse process involves capturing the feature activation maps $f_{t_1}$ and $f_{t_2}$ across multiple output layers of $f_\theta$ using $x_{t_1}$ and $x_{t_2}$ as inputs. These activation maps then undergo a sequence of parameter-sharing DRRE modules to derive image features aligned with the resolution of the input SAR image. By constraining the similarity of these features, it effectively mitigates the impact of various noise components on SAR image terrain features, thereby ensuring both consistency and discrimination of the features. The feature similarity loss $\mathcal{L}_{\text{sim}}$ is defined as follows:

$$\mathcal{L}_{\text{sim}} = \mathbb{E}\left[\left\|\text{DRRE}\left(f_{t_1}\right) - \text{DRRE}\left(f_{t_2}\right)\right\|^2\right]. \tag{10}$$

In order not to interfere with the training of the diffusion model, the gradient of $\mathcal{L}_{\text{sim}}$ is not passed back to $f_\theta$.

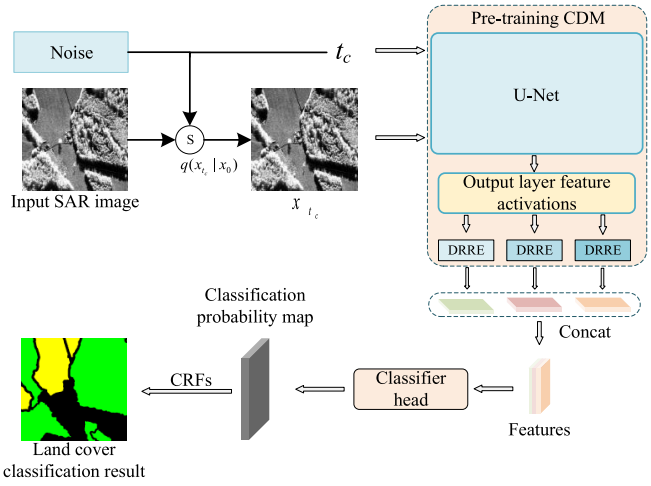



Fig. 4. Pipline of the land cover classification.

In summary, the loss function of the pretrained CDM is defined as follows:

$$\mathcal{L}_{\text{CDM}} = \mathcal{L}_{\text{hybrid}} + \mathcal{L}_{\text{sim}}. \tag{11}$$

### C. Land Cover Classification

The land cover classification task is fine-tuned based on the pretrained CDM with limited labeled sample at the pixel level, which is shown in Fig. 4. For a given input image $x \in R^{H\times W\times 3}$, the noisy image $x_{t_c}$ at classification step $t_c$ is obtained by adding Gaussian noise through 3. The pretrained

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**Algorithm 1** Training Process of CDM

**Input:**

SAR iamges $x$, limited SAR image labels $y_x$, CDM, classification step $t_c$, and classification model $P$.

**Output:**

$P$.

**Pre-training CDM:**
Initializing CDM parameters
**for** number of iterations **do**
  Sample time steps $t_1$ and $t_2$ uniformly from the range $(1, \ldots, T)$
  $\epsilon \in \mathcal{N}(0, 1)$
  Compute $x_{t_1}$ and $x_{t_2}$ using formula 3
  Forward computation of $f_\theta(x_{t_1}, t_1)$, $f_\theta(x_{t_2}, t_2)$ and feature activations in the output layers
  The loss function is calculated according to formula 11
  Backpropagation and update the parameter weights of the CDM
**end for**
**Land cover classification:**
Initializing the $P$ classification header
Transfer the weights from the pre-trained CDM to the $P$
**for** epochs **do**
  Forward diffusion to obtain the $x_{t_c}$
  Compute $P(x_{t_c}, t_c)$ and compute $\mathcal{L}_P$ according to formula 12
  Fine-tune the full parameter weights of $P$
**end for**

CDM is used to capture feature activations across multiple output layers and derive an image feature representation of $x$ using a sequence of pretrained DRRE. And the classification probability distribution map is obtained by additional classifier head to achieve SAR image land cover classification. The full parameter weights of the land cover classification model $P$, consisting of the pretrained CDM and the classifier heads, are fine-tuned using a small number of labeled samples. The model's objective function is defined as follows:

$$\mathcal{L}_P = -\frac{1}{H \times W} \sum_{i=1}^{H \times W} \sum_{c=1}^{C} y_x^{i,(c)} \log P\left(x_{t_c}, t\right)^{i,(c)} \tag{12}$$

where $C$ is the number of categories and $y_x$ is the label of $x$. $P(\cdot)$ is the probability distribution of categories output by $P$. Algorithm 1 is the complete training process for the CDM.

In SAR imagery, inherent noise and variability in land cover characteristics often lead to classification errors, particularly along the boundaries of distinct land cover types. Conditional random fields (CRFs) [39] address these challenges by modeling the spatial dependencies between pixels, leveraging both local and global contextual information to refine boundary region classifications and achieve substantial improvements in accuracy. By associating local pixel values with the semantic features of their neighboring regions, CRFs enforce semantic consistency across adjacent pixel labels, thereby eliminating isolated misclassifications and enhancing the overall spatial coherence of the results. Furthermore, the inherent noise characteristics of SAR imagery, coupled with the complexity of imaging conditions such as variations in resolution and polarization modes exacerbate classification challenges. CRFs effectively mitigate these issues by integrating contextual priors into the classification process, thereby suppressing the influence of noise and significantly improving model performance in complex and heterogeneous scenarios. In the testing phase, in order to smooth the prediction label by minimizing clutter points, the output prediction results from $P$ are postprocessed using conditional random fields (CRFs) to generate the final prediction result.

## IV. Experiments

This section presents the results generated by the CDM and compares and analyzes the land cover classification results of the proposed method with several other methods. The datasets used for the experiments are described in Section IV-A. Section IV-B elaborates on the architecture and training process of the pretrained CDM, examining its generation capabilities through visual analysis. The proposed method is compared with several land cover classification methods in Section IV-C. Ablation study and discussion are conducted in Sections IV-D and IV-F, respectively. In addition, we also supplemented the results of the training data at different scales which is shown in Section IV-E.

### A. Experimental Data and Evaluation Indices

In this article, 18 broad high-resolution SAR data were used for experimental analysis. Table I lists the details of these data. The dataset covers different geographical areas, both domestic and international, acquired by several sensors, including China's GF-3. These datasets contain differentiated information on imaging parameters such as resolution, polarization, and band. They provide a rich and diverse terrain representation that enhances the generality of the model. All of the above SAR data were cropped into 256 × 256 image patches and the number of samples contained in each scene is also recorded in Table I. The SAR data are manually annotated, with the exception of the three SAR datasets from Xinxiang, Zhengzhou, and Jiaozuo. The entire SAR dataset will be utilized for CDM pretraining, and it is important to note that no labeling information will be used during this pretraining phase. For land cover classification, only a small amount of labeled data was employed for model fine-tuning, while the remaining labeled information was exclusively utilized during the testing phase to compute quantitative indices for classification.

In this section, land cover classification experiments are carried out on six scenes, Traunstein, PoDelta, Rosenheim[2], Mississippi (north), Mississippi (south), and Pohang, and compared qualitatively and quantitatively with a variety of classification methods. The six test scenes are selected for their similarity in resolution, satellites, and spectral bands to the pretraining scenes, ensuring consistency in imaging conditions. However, they feature different types of land cover and imaging scenarios to evaluate the ability of the model to generalize the features learned during pretraining to diverse real-world applications. This selection allows for testing the robustness of the method and performance across a variety

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TABLE I
INFORMATION OF SAR DATASETS

| Area | Time | Sensor | Size | Resolution | Band | Polarization | Classes | Number of samples |
|---|---|---|---|---|---|---|---|---|
| **Traunstein** | **2007-11-05** | **F-SAR** | **4278 × 6187** | **1.0m × 0.67m** | **X** | **HH** | **$4^1$** | **384** |
| Napoli | 2013-03-01 | Cosmo-SkyMed | 18332 × 16000 | 2.5m | X | HH | $4^1$ | 3738 |
| **PoDelta** | **2007-09-27** | **Cosmo-SkyMed** | **18308 × 16716** | **2.5m** | **X** | **HH** | **$4^1$** | **3657** |
| Istanbul | 2014-08-14 | TerraSAR-X | 26606 × 17296 | 0.3m | X | VV | $4^2$ | 5423 |
| Rosenheim[1] | 2008-01-27 | TerraSAR-X | 7691 × 7224 | 1.75m | X | HH | $4^1$ | 672 |
| Washington | 2013-12-10 | TerraSAR-X | 25454 × 13836 | 0.3m | X | HH | 6 | 3368 |
| Hong Kong Airport | 2009-12-27 | TerraSAR-X | 16309 × 9281 | 1.75m | X | HH | 9 | 1733 |
| **Rosenheim[2]** | **2008-01-27** | **TerraSAR-X** | **3600 × 3600** | **3m** | **X** | **HH** | **$4^1$** | **196** |
| Mississippi(middle) | 2008-02-01 | TerraSAR-X | 16125 × 15166 | 3m | X | HH | $4^3$ | 2539 |
| **Mississippi(north)** | **2008-02-01** | **TerraSAR-X** | **21500 × 15333** | **3m** | **X** | **HH** | **$4^3$** | **3276** |
| **Mississippi(south)** | **2008-02-01** | **TerraSAR-X** | **21666 × 15166** | **3m** | **X** | **HH** | **3** | **3275** |
| Jiujiang | 2016-11-24 | GF-3 | 8000 × 8000 | 3m | C | DV | $4^1$ | 961 |
| Dongying | 2017-04-16 | GF-3 | 9216 × 10240 | 1m | C | VV | 5 | 1440 |
| **Pohang** | **2018-07-13** | **GF-3** | **7680 × 9728** | **1m** | **C** | **HH** | **5** | **1140** |
| Xi'an | 2018-07-28 | GF-3 | 2304 × 3840 | 1m | C | VV | 5 | 135 |
| Xinxiang | 2021-07-15 | GF-3 | 29777 × 27414 | 10m | C | HV | - | 9087 |
| Zhengzhou | 2021-07-22 | GF-3 | 12009 × 21282 | 10m | C | HV | - | 3310 |
| Jiaozuo | 2021-07-20 | GF-3 | 28888 × 26317 | 10m | C | HV | - | 8685 |

* $4^1$ represent four terrain: building, water, farmland and forest.
* $4^2$ represent four terrain: building, water, road and forest.
* $4^3$ represent four terrain: building, water, farmland and algae.
* 5 represent five terrain: building, water, farmland, forest and road.
* 3 represent three terrain: water, marsh and algae.
* 6 represent six terrain: water, forest, high building, barely, low building and road.
* 9 represent nine terrain: water, forest, high building, runway, barely, low building, green and road.
* The data used for testing are highlighted in bold in the table.

of remote sensing image analysis tasks. The selected scenes include all the sensors listed in Table I and cover a wide range of bands, polarizations, and resolution to effectively demonstrate the generalizability of the proposed method. In addition to typical terrains such as water, forests, buildings, and farmland, certain scenes also include terrains like roads, marshes, and algae. These diverse elements contribute to elucidating the discriminative capability of the proposed method in capturing features. The color of each terrain is shown below the corresponding classification result. During fine-tuning and testing, any terrains not manually labeled will be categorized as other. They may introduce confusion and is not conducive to precise feature analysis, thus will be ignored. During the preprocessing, similar to our prior study [28], we transformed the raw 16-bit SAR data into 8-bit data using truncated linear stretching to facilitate efficient processing with standard DL networks. Specifically, we analyzed the distribution of gray-level proportions in the 16-bit SAR data and find that most pixel values are concentrated at low gray levels, while high gray values accounted for only a small fraction of pixels. So a linear mapping approach is employed with a 95th-percentile threshold truncation, effectively compressing the dynamic range. This method minimizes the compression of most pixel values into a narrow range caused by a small number of excessively high gray values, thereby preserving the dynamic range of SAR images with minimal loss on pixel resolution.

Four overall evaluation indices were used to measure land cover classification results, namely overall accuracy (OA), kappa coefficient, mean intersection over union (MIoU), and frequency weight intersection over union (FWIoU). OA quantifies the percentage of accurately classified pixels, while the kappa coefficient gauges agreement between ground truth and predicted classification maps. MIoU calculates the ratio of the intersection to the union of true and predicted values. FWIoU, a variant of MIoU, applies weights based on category occurrence frequencies. In addition to the four overall indices mentioned earlier, $F1$ scores, a harmonized average of precision and recall, were employed in the experiments to evaluate the classification performance across various terrains.

All experiments in this article are implemented on the framework of PyTorch 1.7.1 with four NVIDIA RTX 2080Ti GPUs.

### B. Generation Analysis of CDM

*1) Settings:* The CDM employs a U-Net network structure with self-attention [38]. Feature activations from layers 4, 6, 8, and 10 of the output layer are fed into corresponding DRRE modules to derive an SAR image feature representation. The total step $T$ is set to 1000 with learnable variance and cos noise scheduling. The training process uses an exponential moving average with weights of 0.9999. The learning rate is initially $1e^{-4}$, and a linear descent strategy is used, and batch size is 8. The number of iterations is 2 50 000 and the optimizer

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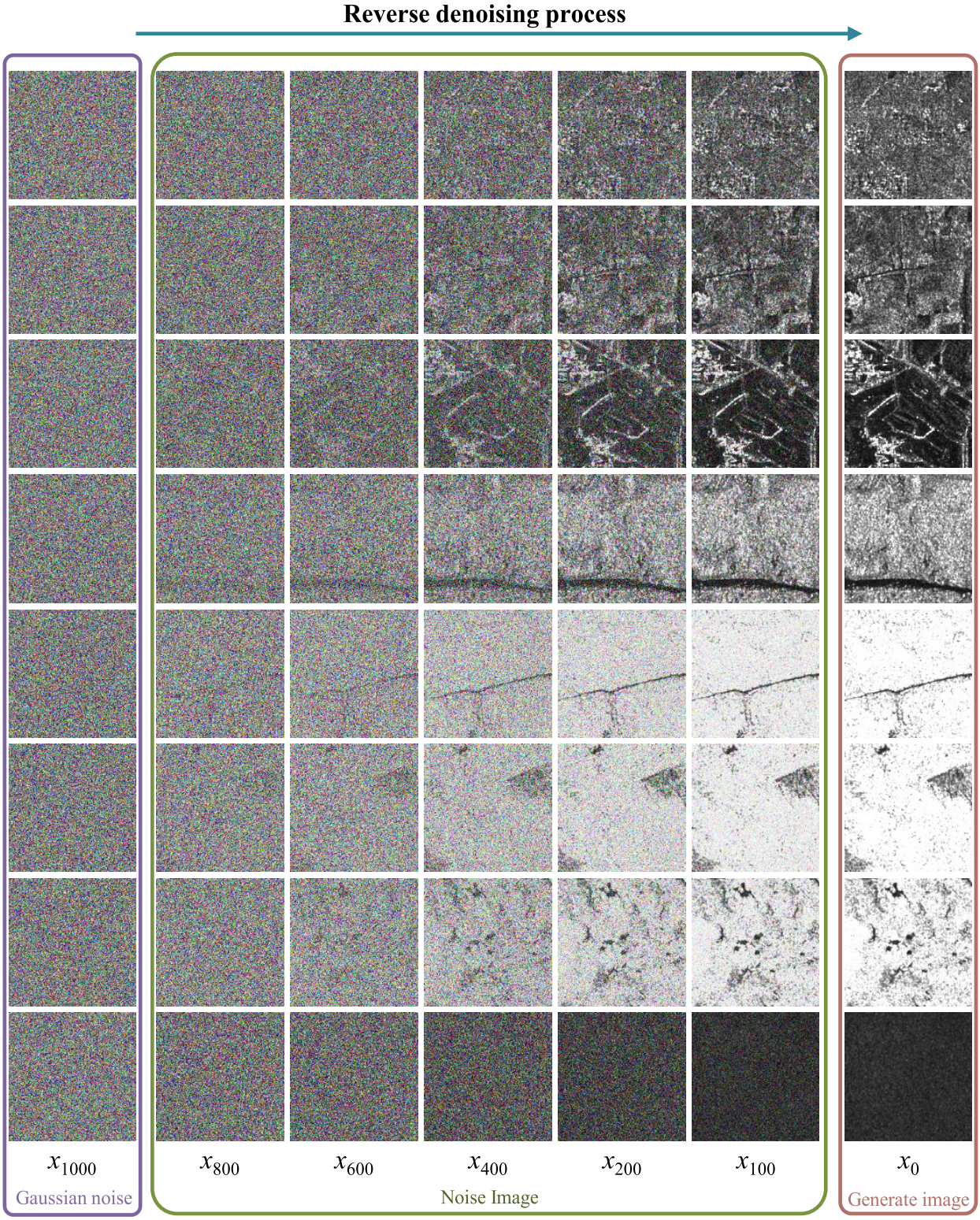



Fig. 5. Visualization of the reverse denoising process for the CDM.

is a weightless decaying AdamW. The selection of the above parameters mainly refers to [31] and [38]. The pretraining dataset contains all the samples in Table I, a total of 53 019 SAR images of 256 × 256 size.

*2) Experiments and Analyses:* To demonstrate the generative effect of the pretrained diffusion model, Fig. 5 shows the intermediate images generated by the CDM over several steps of the reverse denoising process. Starting from a random Gaussian noise, the generated image exhibits discernible terrain and distinct brightness values for different scattering types after 500 steps of the reverse denoising process. As the denoising progresses, the intensity of the Gaussian noise in the image diminishes significantly, revealing clearer terrain features. The final generated image $x_0$ exhibits exceptional

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TABLE II
MULTI-INDICATOR DENOISING PERFORMANCE EVALUATION

| Metric | Step | SAR Dataset | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | **SAR1** | **SAR2** | **SAR3** | **SAR4** | **SAR5** | **SAR6** | **SAR7** | **SAR8** |
| PSNR ↑ | 0 | 7.795 | 7.550 | 6.349 | 6.374 | 7.422 | 4.627 | 4.769 | 5.333 |
| | 100 | 8.008 | 7.849 | 6.781 | 6.839 | 7.726 | 5.258 | 5.389 | 5.871 |
| | 200 | 8.330 | 8.201 | 7.307 | 7.359 | 8.105 | 5.976 | 6.102 | 6.570 |
| | 400 | 9.348 | 9.312 | 8.836 | 8.925 | 9.243 | 8.035 | 8.142 | 8.412 |
| | 600 | 11.418 | 11.437 | 11.494 | 11.668 | 11.431 | 11.444 | 11.479 | 11.513 |
| | 800 | 16.170 | 16.194 | 16.670 | 16.692 | 16.318 | 17.590 | 17.484 | 17.164 |
| | 1000 | **49.938** | **49.924** | **49.897** | **49.649** | **49.888** | **50.778** | **50.729** | **50.260** |
| SSIM ↑ | 0 | 0.005 | 0.006 | 0.006 | 0.004 | 0.005 | 0.005 | 0.006 | 0.006 |
| | 100 | 0.030 | 0.032 | 0.028 | 0.007 | 0.034 | 0.017 | 0.017 | 0.025 |
| | 200 | 0.060 | 0.059 | 0.052 | 0.009 | 0.069 | 0.027 | 0.029 | 0.049 |
| | 400 | 0.133 | 0.132 | 0.120 | 0.016 | 0.149 | 0.063 | 0.063 | 0.113 |
| | 600 | 0.265 | 0.257 | 0.234 | 0.037 | 0.295 | 0.135 | 0.136 | 0.235 |
| | 800 | 0.551 | 0.539 | 0.465 | 0.122 | 0.602 | 0.341 | 0.347 | 0.516 |
| | 1000 | **1.000** | **1.000** | **0.999** | **0.996** | **1.000** | **0.998** | **0.998** | **0.999** |
| IFC ↑ | 0 | 0.832 | 0.815 | 0.695 | 0.883 | 0.848 | 0.893 | 0.891 | 0.883 |
| | 100 | 0.841 | 0.826 | 0.714 | 0.873 | 0.866 | 0.912 | 0.910 | 0.901 |
| | 200 | 0.853 | 0.839 | 0.738 | 0.866 | 0.884 | 0.929 | 0.926 | 0.920 |
| | 400 | 0.882 | 0.872 | 0.800 | 0.854 | 0.918 | 0.960 | 0.959 | 0.953 |
| | 600 | 0.923 | 0.917 | 0.881 | 0.859 | 0.955 | 0.984 | 0.983 | 0.979 |
| | 800 | 0.972 | 0.970 | 0.961 | 0.909 | 0.986 | 0.996 | 0.996 | 0.994 |
| | 1000 | **1.000** | **1.000** | **1.000** | **1.000** | **1.000** | **1.000** | **1.000** | **1.000** |
| MAE ↓ | 0 | 87.755 | 89.727 | 100.410 | 100.405 | 90.616 | 121.315 | 119.130 | 111.736 |
| | 100 | 85.516 | 86.383 | 94.425 | 93.953 | 87.266 | 109.215 | 107.690 | 102.827 |
| | 200 | 82.090 | 82.577 | 87.890 | 87.519 | 83.075 | 97.358 | 96.411 | 92.667 |
| | 400 | 72.091 | 71.782 | 72.215 | 71.686 | 71.937 | 72.116 | 71.817 | 72.014 |
| | 600 | 55.827 | 55.467 | 52.905 | 52.456 | 54.959 | 46.255 | 46.638 | 49.032 |
| | 800 | 31.762 | 31.694 | 29.701 | 30.303 | 30.867 | 22.206 | 22.732 | 25.443 |
| | 1000 | **0.561** | **0.562** | **0.563** | **0.586** | **0.564** | **0.478** | **0.478** | **0.532** |
| LPIPS ↓ | 0 | 0.727 | 0.763 | 0.769 | 0.761 | 0.755 | 0.745 | 0.759 | 0.742 |
| | 100 | 0.726 | 0.752 | 0.761 | 0.758 | 0.754 | 0.748 | 0.758 | 0.736 |
| | 200 | 0.719 | 0.741 | 0.752 | 0.756 | 0.744 | 0.738 | 0.756 | 0.726 |
| | 400 | 0.704 | 0.720 | 0.718 | 0.746 | 0.727 | 0.706 | 0.736 | 0.681 |
| | 600 | 0.639 | 0.652 | 0.615 | 0.708 | 0.658 | 0.648 | 0.667 | 0.590 |
| | 800 | 0.458 | 0.466 | 0.397 | 0.625 | 0.478 | 0.540 | 0.546 | 0.435 |
| | 1000 | **0.001** | **0.001** | **0.002** | **0.011** | **0.001** | **0.002** | **0.002** | **0.001** |
| DISTS ↓ | 0 | 0.502 | 0.519 | 0.584 | 0.515 | 0.519 | 0.521 | 0.536 | 0.538 |
| | 100 | 0.500 | 0.517 | 0.574 | 0.508 | 0.518 | 0.518 | 0.528 | 0.528 |
| | 200 | 0.487 | 0.502 | 0.559 | 0.498 | 0.506 | 0.512 | 0.524 | 0.516 |
| | 400 | 0.464 | 0.474 | 0.519 | 0.481 | 0.481 | 0.478 | 0.487 | 0.469 |
| | 600 | 0.415 | 0.414 | 0.437 | 0.454 | 0.426 | 0.410 | 0.409 | 0.391 |
| | 800 | 0.302 | 0.307 | 0.280 | 0.402 | 0.310 | 0.295 | 0.297 | 0.293 |
| | 1000 | **0.002** | **0.002** | **0.003** | **0.012** | **0.001** | **0.002** | **0.001** | **0.002** |

* ↑ means the larger the value,the higher the similarity, ↓ means the lower the value, the higher the similarity

quality in terms of pixel value distribution and image terrain characteristics. Moreover, it demonstrates the ability to learn and reproduce the inherent coherent speckle noise typical of SAR images to a significant extent. Through the quantitative analysis of the model's denoising effectiveness in Table II, by examining the trends of key metrics such as PSNR, SSIM, IFC, MAE, LPIPS, and DISTS across training iterations, it is evident that the model demonstrates exceptional performance

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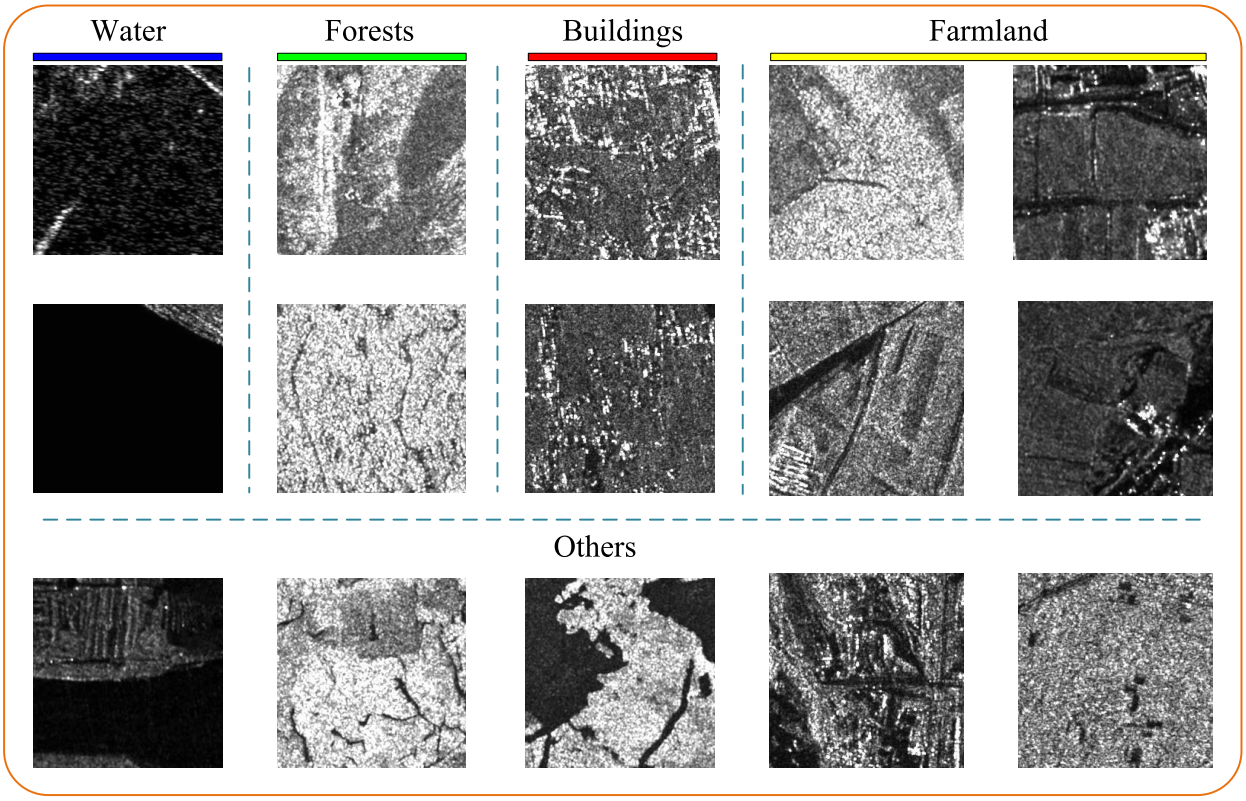



Fig. 6. CDM generates image visualizations.

in noise reduction tasks. Specifically, as the training progresses, a notable increase is observed in PSNR and SSIM values, which indicates significant improvements in the quality and structural similarity of processed images. Concurrently, a downward trend in MAE, LPIPS, and DISTS values indicates that the discrepancies between the model's predictions and the ground truth decrease progressively, with the generated images approaching perceptual fidelity to the original ones. Moreover, the enhancement in IFC values corroborates the enhanced fidelity of information. Collectively, these observations demonstrate that the model effectively achieves noise removal while maintaining or even enhancing the image quality and informational consistency. This demonstrates the effective ability of the pretrained CDM to decouple coherent speckle noise and terrain from extensive, unlabeled SAR data, and to capture essential semantic information within the images. The model exhibits robust feature characterization and exceptional image generation capabilities.

Fig. 6 illustrates the different types of SAR image terrain samples generated by the pretrained CDM. The generated images closely resemble real SAR images, showcasing the CDM's ability to effectively capture, recognize, and differentiate various terrains. The experimental results above demonstrate that the pretrained CDM can generate discriminative SAR image features for land cover classification, thereby enhancing the accuracy of the classification model. These samples are used solely as a reference to demonstrate the performance of the model on the upstream task, and do not participate in the training or parameter fine-tuning of downstream tasks.

### C. Comparison With the State-of-the-Art Methods

*1) Settings:* The land cover classification model $P$ consists of a CDM and a classifier head. The parameter weights of the pretraining CDM are fully transferred to $P$, and feature activations from the DRRE output are spliced and fed to the classifier head to obtain the probabilistic map of the terrain distribution of SAR images. The classifier head consists of two convolutional layers of size $3 \times 3$, with a stride of 1, 1 pixel padding, followed by a ReLU activation layer and a BN layer.

For fine-tuning, each scene utilizes 16 labeled samples for training and 24 labeled samples for validation, except for the PoDelta scene, which employs a training set of 22 labeled samples and a validation set of 20 labeled samples. The training batch size is 4, with a classification step $t_c$ of 25. We utilize a learning rate of $1e^{-4}$ and employ the Adam optimizer with a weight decay of 0.05. Additionally, all images are standardized to a size of $256 \times 256$.

*2) Experiments and Analyses:* Four classification methods, DeepLabv3+ [40], CCNR [41], SegFormer [42], and MAE [37], are compared with the proposed CDM in the land cover classification experiments. DeepLabv3+ and SegFormer represent typical approaches to semantic segmentation, utilizing CNN and transformer, respectively. MAE represents a cutting-edge approach to generative self-supervised learning, relying on masked autoencoders. CCNR is an SAR image land cover classification method using cross-region context and noise regularization. Baseline is a semantic segmentation method that builds on DDPM, leveraging pretrained DDPM feature

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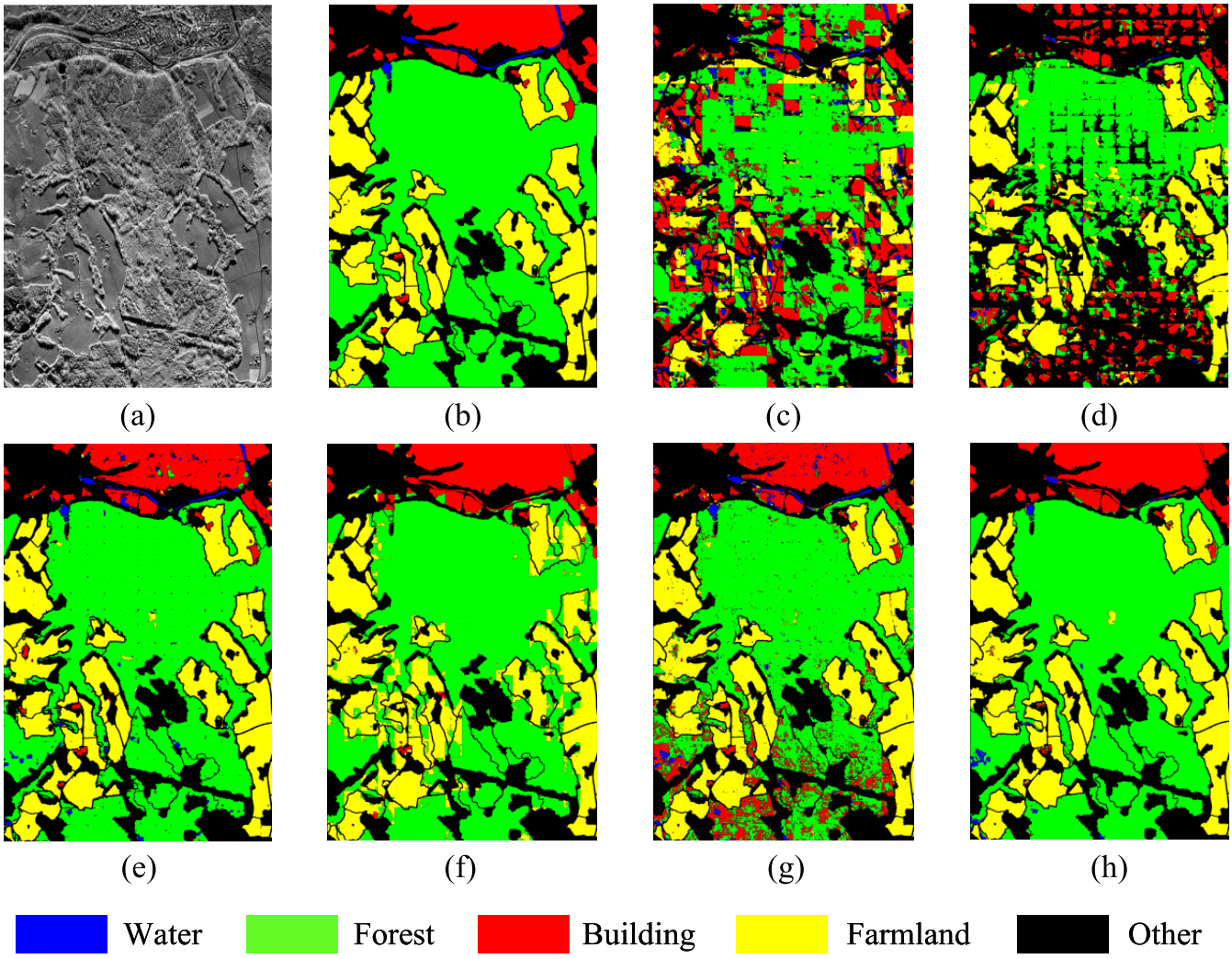



Fig. 7. Classification maps of different methods in Traunstein. (a) SAR image. (b) Ground Truth. (c) DeepLabv3+. (d) CCNR. (e) SegFormer. (f) MAE. (g) Baseline (DDPM). (h) CDM.

activations from multiple output layers as image features. It employs a lightweight CNN network as a classification head for land cover classification. The pretraining dataset used by Baseline aligns with that of CDM. For the comparison experiments, the backbone of DeepLabv3+ is ResNet101 [43]. Baseline uses feature activations from output layers 4, 5, 6, 7, 8, 9, and 10, where the step size is 25, to derive image features. Among the methods mentioned above, DeepLabv3+, CCNR, and SegFormer are supervised methods, and the rest of the methods are self-supervised methods. The full weighting parameters for self-supervised methods were fine-tuned.

*a) Results of Traunstein:* Traunstein is a high-resolution scene characterized by rich texture details and clear terrain edges, which generally simplifies land cover classification. However, due to the scarcity of water samples in this scene, with most being narrow rivers, water identification proves to be more challenging. Fig. 7 shows the prediction maps for different classification methods. DeepLabv3+ and CCNR have significant blocking artifacts and struggle to effectively discriminate between different types of terrain. This is due to overfitting caused by limited labeled samples. MAE has strong terrain detection capabilities, except for water. In addition, MAE exhibits limitations in detecting edges of various types of terrain, which may be attributed to the use of a large-scale mask that reduces the model's ability to perceive fine edge details. SegFormer and Baseline demonstrate strong water perception. However, they still exhibit isolated misclassified clutter spots. Moreover, the clutter in the Baseline is worse because of the need to introduce noise information. The CDM effectively reduces a substantial number of the clutter through CRFs, but also result in the omission of the water.

The quantitative indicators of the different methods on Traunstein are given in Table III. The SegFormer is most sensitive over water, thanks to its advanced model structure. Compared to SegFormer, CDM has an improvement of approximately 1%, 2%, and 1% on the remaining three terrains, respectively. SegFormer has the highest accuracy on MIoU due to its high water recognition. However, CDM shows 2.3%, 3.8%, and 1.9% improvement over SegFormer on OA, kappa, and FWIoU, respectively, and 8.7%, 14.4%, 11.3%, and 13.3% improvement over Baseline on the overall indices.

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TABLE III
CLASSIFICATION RESULTS ON TRAUNSTEIN

| Method | F1 Score | | | | OA | kappa | MIoU | FWIoU |
|---|---|---|---|---|---|---|---|---|
| | Water | Forest | Building | Farmland | | | | |
| Deeplabv3+ | 0.174 | 0.743 | 0.279 | 0.601 | 0.588 | 0.355 | 0.320 | 0.494 |
| CCNR | 0.196 | 0.711 | 0.454 | 0.895 | 0.639 | 0.483 | 0.441 | 0.592 |
| Segformer | **0.555** | 0.979 | 0.944 | 0.975 | 0.957 | 0.925 | **0.797** | 0.945 |
| MAE | 0.000 | 0.923 | 0.919 | 0.853 | 0.898 | 0.817 | 0.613 | 0.818 |
| Baseline (DDPM) | 0.419 | 0.914 | 0.717 | 0.965 | 0.893 | 0.819 | 0.649 | 0.831 |
| CDM | 0.290 | **0.988** | **0.966** | **0.984** | **0.980** | **0.963** | 0.762 | **0.964** |

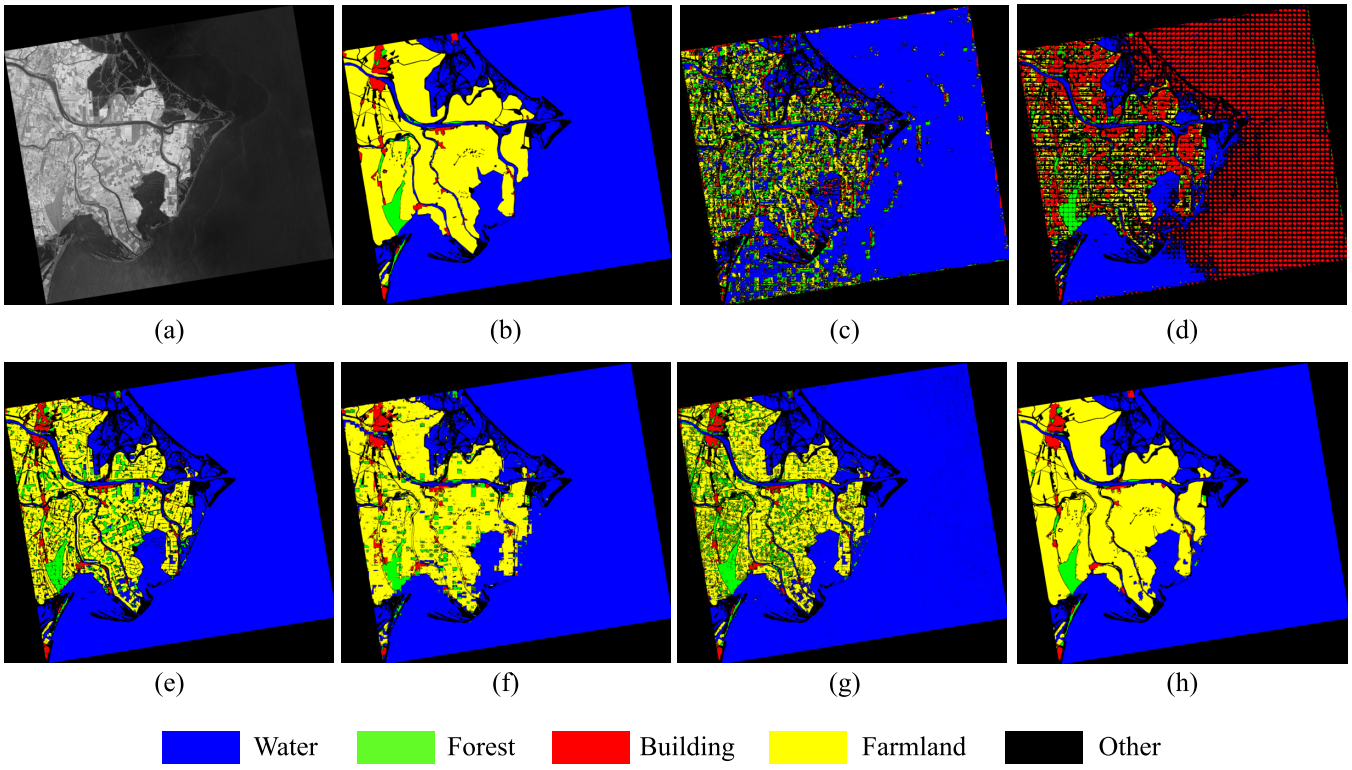



Fig. 8. Classification maps of different methods in PoDelta. (a) SAR image. (b) Ground Truth. (c) DeepLabv3+. (d) CCNR. (e) SegFormer. (f) MAE. (g) Baseline (DDPM). (h) CDM.

*b) Results of PoDelta:* Fig. 8 shows the classification results of different methods in PoDelta. In contrast to the terrain distribution in Traunstein, the PoDelta exhibits a diverse range of water bodies and smaller rivers. The abundance of water samples and distinct differences in terrain brightness make identification easier. Although most farmland regions are distinctly different from forests, there exists a small portion of farmland with terrain characteristics that closely resemble forests. Relying solely on texture or brightness features makes it challenging for the model to effectively distinguish this part of the terrain. Furthermore, distinguishing between forests and buildings in PoDelta scenes is challenging due to blurred differences, posing a significant challenge for the model's recognition capabilities. DeepLabv3+ and CCNR are not able to effectively complete the generalization of the model with a small number of samples to identify different types of terrain. SegFormer utilizes a global attention mechanism to learn contextual features across the entire input image, allowing it to accurately identify most types of terrain. However, it has difficulty distinguishing between farmland and forests with very similar features. Building on extensive pretraining with large-scale samples, both MAE and Baseline demonstrate effective differentiation of various terrains. However, they exhibit considerable misclassification clutter, particularly Baseline, which needs to introduce noise information. CDM correctly identifies almost all terrains and is very close to manual labeling.

The quantitative indicators of the different methods on PoDelta are given in Table IV. CDM has the highest accuracy in all terrain evaluation indicators. Its OA reaches 0.978, which is 27.9%, 72.3%, 8.0%, 4.4%, and 6.4% higher than other methods. This clearly illustrates the validity of the proposed land cover classification method for SAR images.

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TABLE IV
CLASSIFICATION RESULTS ON PODELTA

| Method | F1 Score | | | | OA | kappa | MIoU | FWIoU |
|---|---|---|---|---|---|---|---|---|
| | Water | Forest | Building | Farmland | | | | |
| Deeplabv3+ | 0.876 | 0.077 | 0.128 | 0.512 | 0.699 | 0.421 | 0.308 | 0.635 |
| CCNR | 0.401 | 0.383 | 0.058 | 0.360 | 0.255 | 0.128 | 0.184 | 0.238 |
| Segformer | 0.982 | 0.346 | 0.687 | 0.825 | 0.898 | 0.783 | 0.600 | 0.873 |
| MAE | 0.975 | 0.374 | 0.593 | 0.894 | 0.934 | 0.855 | 0.603 | 0.894 |
| Baseline (DDPM) | 0.982 | 0.282 | 0.656 | 0.845 | 0.914 | 0.813 | 0.587 | 0.882 |
| CDM | **0.989** | **0.864** | **0.843** | **0.964** | **0.978** | **0.952** | **0.850** | **0.958** |

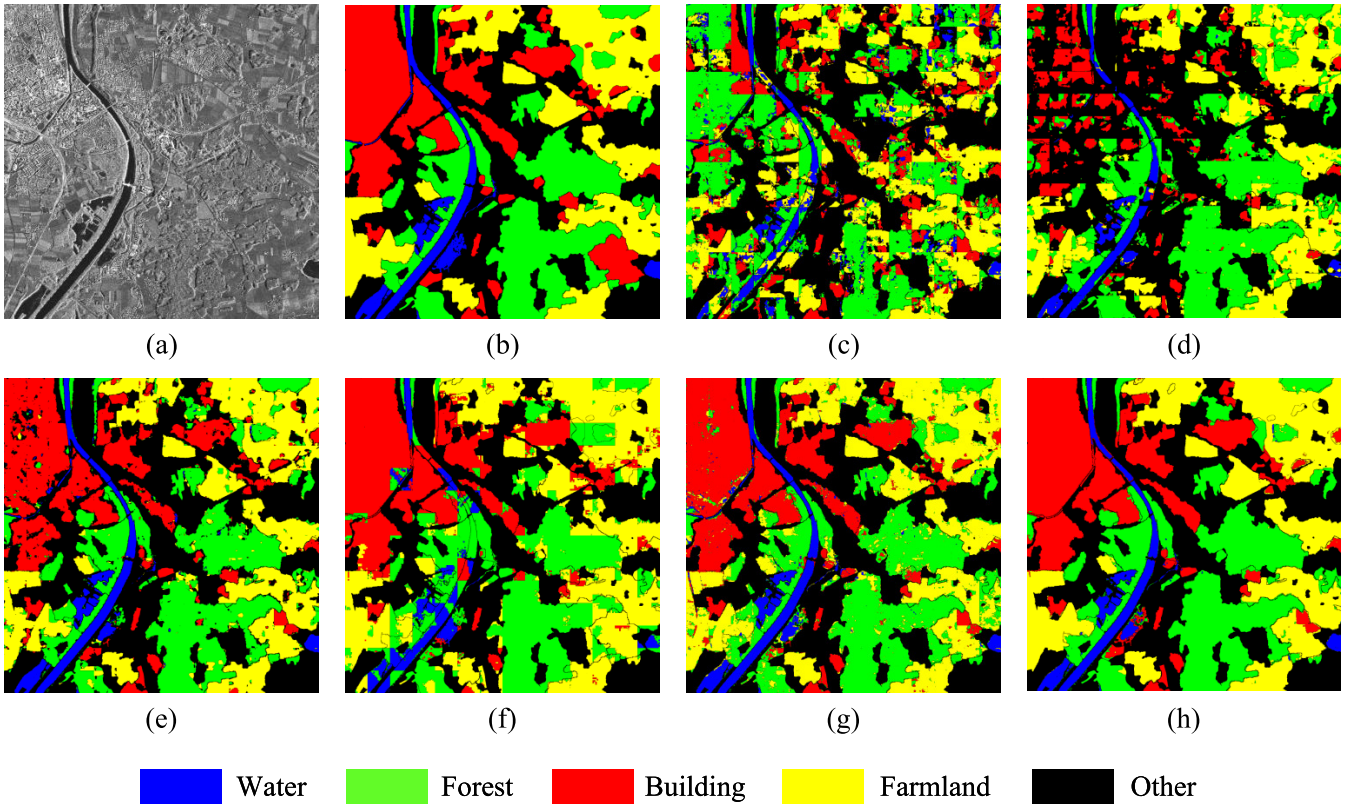



Fig. 9. Classification maps of different methods in Rosenheim$^2$. (a) SAR image. (b) Ground Truth. (c) DeepLabv3+. (d) CCNR. (e) SegFormer. (f) MAE. (g) Baseline (DDPM). (h) CDM.

*c) Results of Rosenheim$^2$:* Fig. 9 shows the classification results of different methods in Rosenheim$^2$. Fig. 9(a) and (b) reveals a mixture of terrain within the scene, characterized by complex terrain boundaries. Effective differentiation of different terrains and their boundaries is therefore key to improving model performance. DeepLabv3+ struggles to learn terrain features effectively when provided with limited labeled samples, resulting in challenges in correctly identifying terrains other than water. CCNR is based on cross-region sensing, which improves the model's sensing ability under limited samples, but there is a serious blocking artifact. MAE mitigates the blocking artifact in pretraining, but is insufficient for terrain edge recognition. SegFormer and Baseline enhance the capacity to detect terrain boundaries. However, the outcomes exhibit increased clutter. With the CRFs, CDM has the best classification results.

The quantitative indicators of the different methods on Rosenheim$^2$ are given in Table V. CDM accuracy on water is consistent with SegForemr and similar to Baseline. CDM outperforms DeepLabv3+, CCNR, SegFormer, MAE, and Baseline on OA, kappa, MIoU, FWIoU, and $F1$ score for the remaining terrain, and has a performance improvement of nearly 6% over Baseline on all four overall indices.

*d) Results of Mississippi (North):* The subtle distinction between farmland and buildings in this scene poses a challenge for recognition. Fig. 10 shows the classification results of different methods in Mississippi (north). The multiscale and atrous convolution structure of DeepLabv3+ does not enhance the model's capability to discern similar terrain with limited samples, resulting in poor classification outcomes. The cross-regional mechanism of CCNR enhances the identification of distinct algae and water, but its usefulness for recognizing

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TABLE V
CLASSIFICATION RESULTS ON ROSENHEIM[2]

| Method | F1 Score | | | | OA | kappa | MIoU | FWIoU |
|---|---|---|---|---|---|---|---|---|
| | Water | Forest | Building | Farmland | | | | |
| Deeplabv3+ | 0.517 | 0.500 | 0.424 | 0.608 | 0.493 | 0.309 | 0.347 | 0.347 |
| CCNR | 0.806 | 0.834 | 0.539 | 0.826 | 0.686 | 0.587 | 0.616 | 0.605 |
| Segformer | **0.906** | 0.915 | 0.882 | 0.886 | 0.867 | 0.816 | 0.815 | 0.812 |
| MAE | 0.624 | 0.790 | 0.824 | 0.795 | 0.792 | 0.703 | 0.617 | 0.656 |
| Baseline (DDPM) | 0.899 | 0.885 | 0.911 | 0.860 | 0.885 | 0.838 | 0.800 | 0.796 |
| CDM | **0.906** | **0.960** | **0.929** | **0.935** | **0.940** | **0.915** | **0.875** | **0.887** |

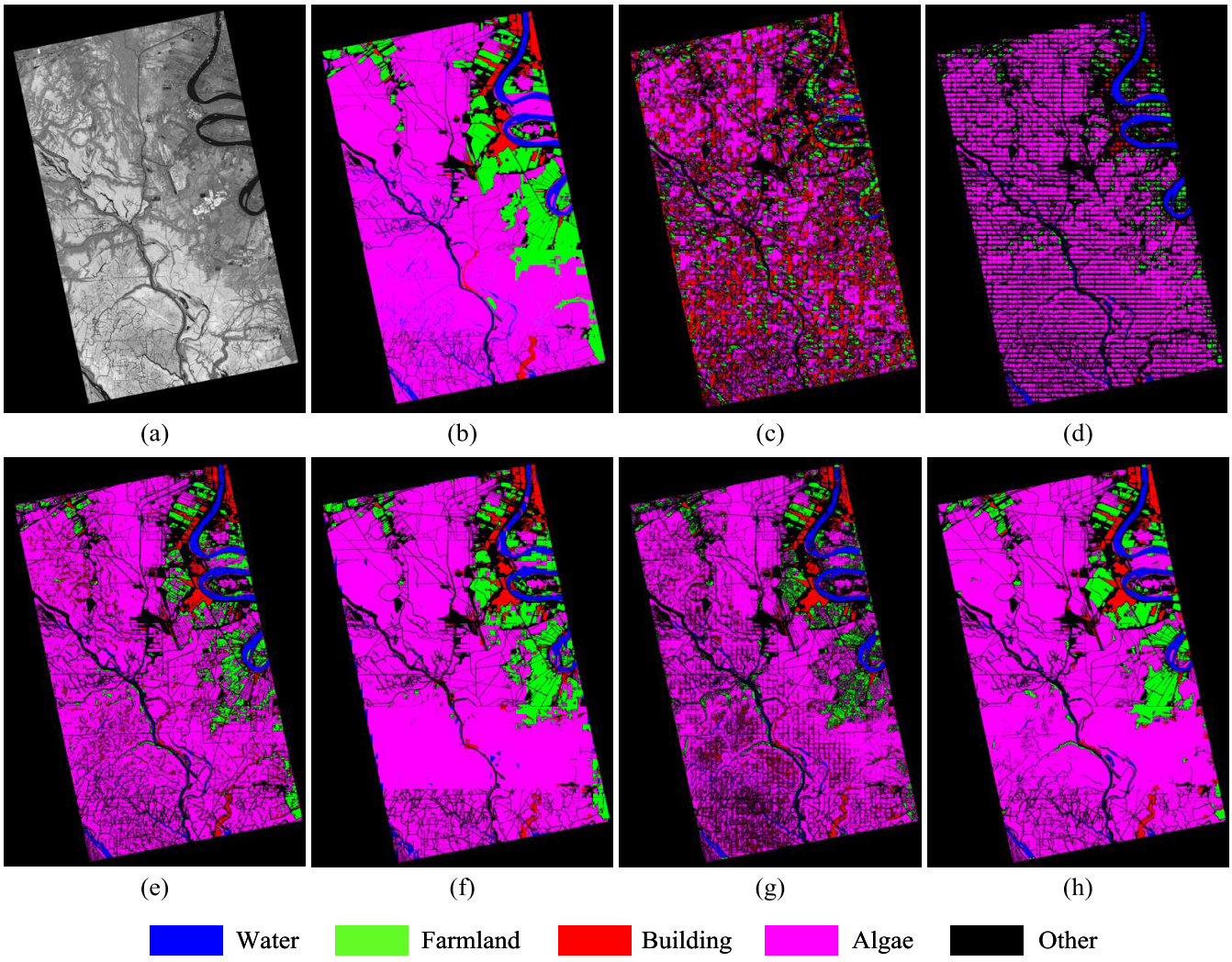



Fig. 10. Classification maps of different methods in Mississippi (north). (a) SAR image. (b) Ground Truth. (c) DeepLabv3+. (d) CCNR. (e) SegFormer. (f) MAE. (g) Baseline (DDPM). (h) CDM.

farmland and buildings is limited. SegFormer and Baseline further enhance the identification of similar terrain. MAE has a strong discriminative feature learning ability with fewer samples due to the generative pretraining process. CDM is more accurate in segmenting buildings and farmland, but there is leakage of some farmland.

The quantitative indicators of the different methods on Mississippi (north) are given in Table VI. The results demonstrate that MAE achieves the highest accuracy in identifying farmland. Moreover, CDM exhibits superior overall performance across various indices, showing a 2% to 4% improvement over the next best method.

*e) Results of Mississippi (South):* The imaging parameters of Mississippi (south) are consistent with those of Mississippi (north), so there is a high degree of similarity between the SAR images. Fig. 11 shows the classification

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TABLE VI
CLASSIFICATION RESULTS ON MISSISSIPPI (NORTH)

| Method | F1 Score | | | | OA | kappa | MIoU | FWIoU |
|---|---|---|---|---|---|---|---|---|
| | Water | Farmland | Building | Algae | | | | |
| Deeplabv3+ | 0.365 | 0.199 | 0.117 | 0.640 | 0.454 | 0.119 | 0.217 | 0.393 |
| CCNR | 0.800 | 0.252 | 0.281 | 0.750 | 0.574 | 0.214 | 0.394 | 0.525 |
| Segformer | 0.840 | 0.565 | 0.565 | 0.898 | 0.812 | 0.543 | 0.582 | 0.737 |
| MAE | 0.547 | **0.745** | 0.703 | 0.938 | 0.887 | 0.684 | 0.599 | 0.804 |
| Baseline (DDPM) | 0.842 | 0.550 | 0.408 | 0.879 | 0.804 | 0.522 | 0.537 | 0.706 |
| CDM | **0.865** | 0.715 | **0.843** | **0.950** | **0.914** | **0.756** | **0.738** | **0.842** |

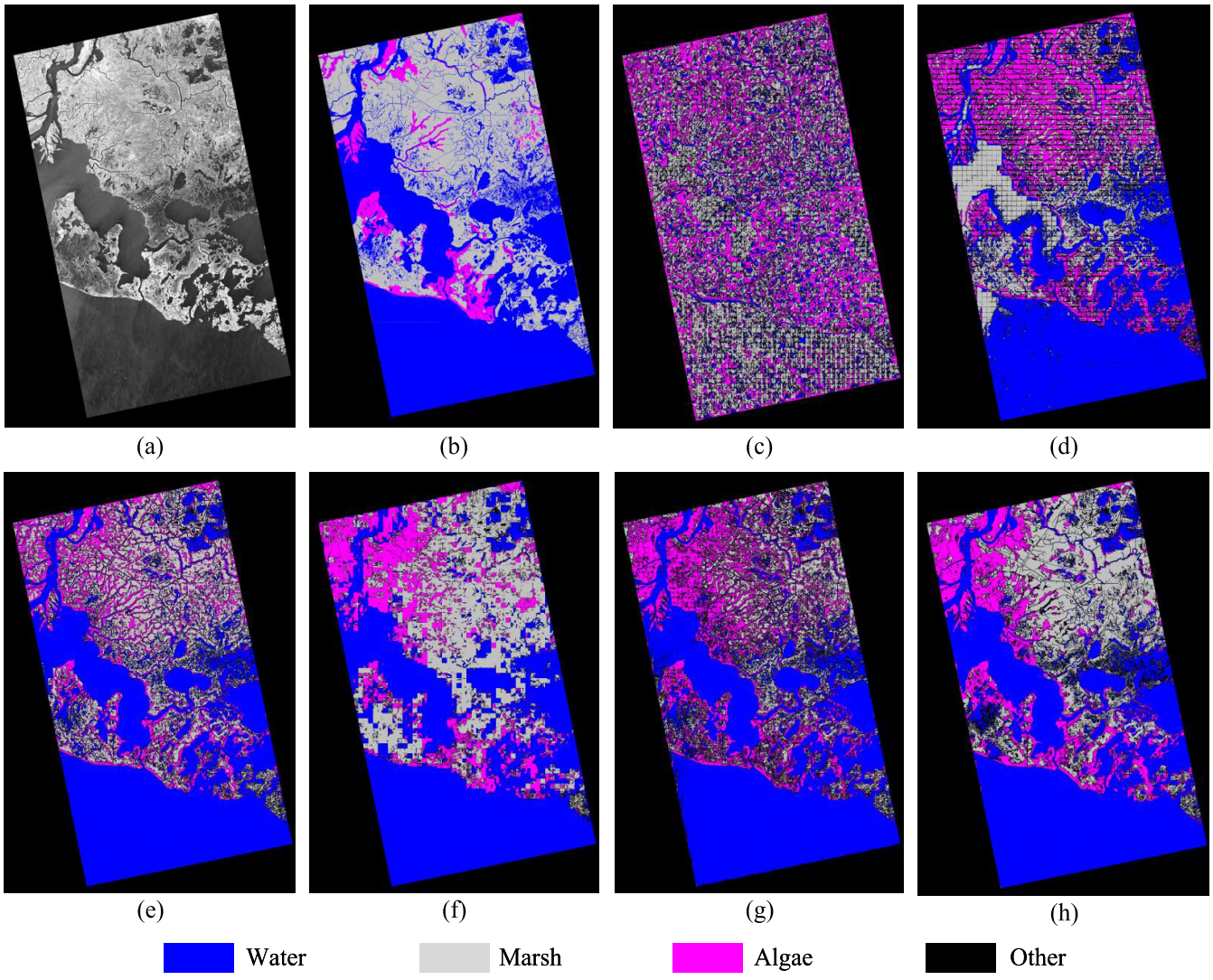



Fig. 11. Classification maps of different methods in Mississippi (south). (a) SAR image. (b) Ground Truth. (c) DeepLabv3+. (d) CCNR. (e) SegFormer. (f) MAE. (g) Baseline (DDPM). (h) CDM.

results of different methods in Mississippi (south). Similar to the visual results from Mississippi (north), the CDM and MAE had similar discrimination on the marsh, but the CDM did not have a significant blocking artifact and therefore had better overall segmentation. In addition, none of the models were able to effectively identify the marsh in the upper left region due to the lack of sufficient contextual information.

The quantitative indicators of the different methods on Mississippi (south) are given in Table VII. The results show that CDM achieves the highest performance across all quantitative evaluation indices. Compared to the SegFormer method, which ranks second, the proposed method demonstrates improvements of 0.3% to 2.0% across all indices.

*f) Results of Pohang:* Unlike the five SAR scenes mentioned above, the Pohang scene contains road terrain with

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TABLE VII
CLASSIFICATION RESULTS ON MISSISSIPPI (SOUTH)

| Method | F1 Score | | | OA | kappa | MIoU | FWIoU |
|---|---|---|---|---|---|---|---|
| | Water | Marsh | Algae | | | | |
| Deeplabv3+ | 0.427 | 0.465 | 0.187 | 0.397 | 0.115 | 0.226 | 0.273 |
| CCNR | 0.896 | 0.533 | 0.331 | 0.669 | 0.479 | 0.458 | 0.592 |
| Segformer | 0.973 | 0.773 | 0.380 | 0.833 | 0.715 | 0.604 | 0.774 |
| MAE | 0.869 | 0.672 | 0.327 | 0.732 | 0.548 | 0.490 | 0.626 |
| Baseline (DDPM) | 0.970 | 0.642 | 0.328 | 0.767 | 0.619 | 0.537 | 0.705 |
| CDM | **0.976** | **0.782** | **0.434** | **0.843** | **0.734** | **0.624** | **0.784** |

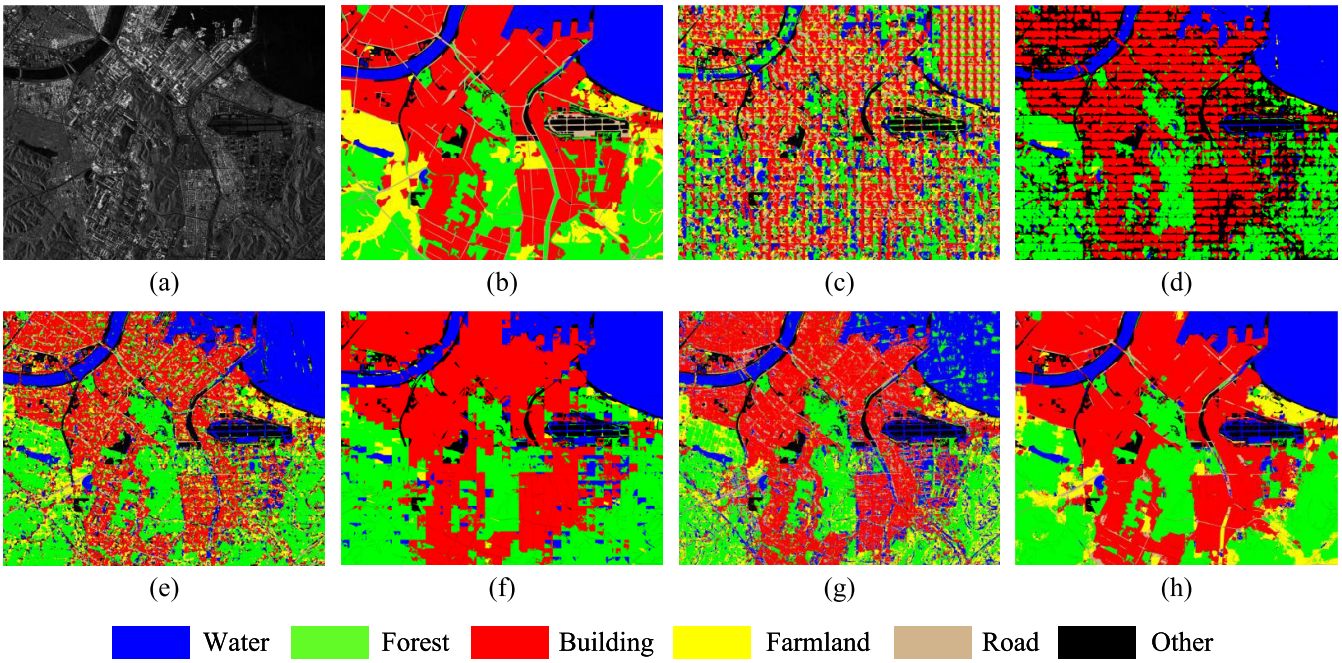



Fig. 12. Classification maps of different methods in Pohang. (a) SAR image. (b) Ground Truth. (c) DeepLabv3+. (d) CCNR. (e) SegFormer. (f) MAE. (g) Baseline (DDPM). (h) CDM.

fewer samples, limited coverage, and a scattering type similar to water. Fig. 12 shows the classification results of different methods in Pohang. Upon comparing the ground truth with the SAR image, it becomes evident that the brightness and texture characteristics of roads closely resemble those of water. Consequently, recognizing roads with limited samples poses a significant challenge, and even with extensive pretraining on large datasets, effectively learning the features of road remains difficult. DeepLabv3+ had severe confounding, while CCNR and MAE were able to recognize building and forest effectively, but with severe blocking artifacts that severely hampered classification performance. SegFormer and Baseline are not able to accurately distinguish even well-characterized waters. The CDM effectively mitigates the blocking artifact and reduces the presence of multiple clutter points, allowing it to detect extended roads within buildings and forests. However, as with all comparison methods, the CDM is still not effective in recognizing and distinguishing large road regions.

The quantitative indicators of the different methods on Pohang are given in Table VIII. The CDM exhibits the highest accuracy in road detection, demonstrating that even with sparse terrain, the proposed method is able to learn to a significant extent, with extracted features exhibiting strong discriminative properties. Additionally, the CDM outperforms all other methods across all indices.

*g) Results of Hong Kong Airport:* The Hong Kong Airport scene contains the largest number of land cover classes, with some categories being sparsely distributed, making it a particularly challenging benchmark for the model. Fig. 13 presents the classification results of different methods over the Hong Kong Airport area. Among the comparison methods, CCNR exhibits severe confusion, misclassifying water as airport runway and vice versa. DeepLabv3+ performs poorly in recognizing buildings (including both high-rise and low-rise buildings). SegFormer fails to accurately identify bare land surrounding the airport runway, unable to capture even the basic structural outlines. MAE misclassifies bare land as trees, while DDPM erroneously identifies large areas of trees as water. Moreover, the results of DeepLabv3, SegFormer, and MAE suffer from severe misclassification, leading to

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TABLE VIII
CLASSIFICATION RESULTS ON POHANG

| Method | F1 Score | | | | | OA | kappa | MIoU | FWIoU |
|---|---|---|---|---|---|---|---|---|---|
| | Building | Water | Farmland | Forest | Road | | | | |
| Deeplabv3+ | 0.545 | 0.211 | 0.161 | 0.367 | 0.110 | 0.353 | 0.150 | 0.172 | 0.249 |
| CCNR | 0.731 | 0.881 | 0.067 | 0.631 | 0.000 | 0.582 | 0.450 | 0.372 | 0.482 |
| SegFormer | 0.705 | 0.825 | 0.379 | 0.646 | 0.257 | 0.622 | 0.502 | 0.421 | 0.492 |
| MAE | 0.844 | 0.828 | 0.108 | 0.654 | 0.000 | 0.714 | 0.585 | 0.396 | 0.547 |
| Baseline (DDPM) | 0.782 | 0.684 | 0.361 | 0.585 | 0.205 | 0.630 | 0.498 | 0.382 | 0.490 |
| CDM | **0.923** | **0.936** | **0.428** | **0.765** | **0.426** | **0.814** | **0.734** | **0.580** | **0.700** |

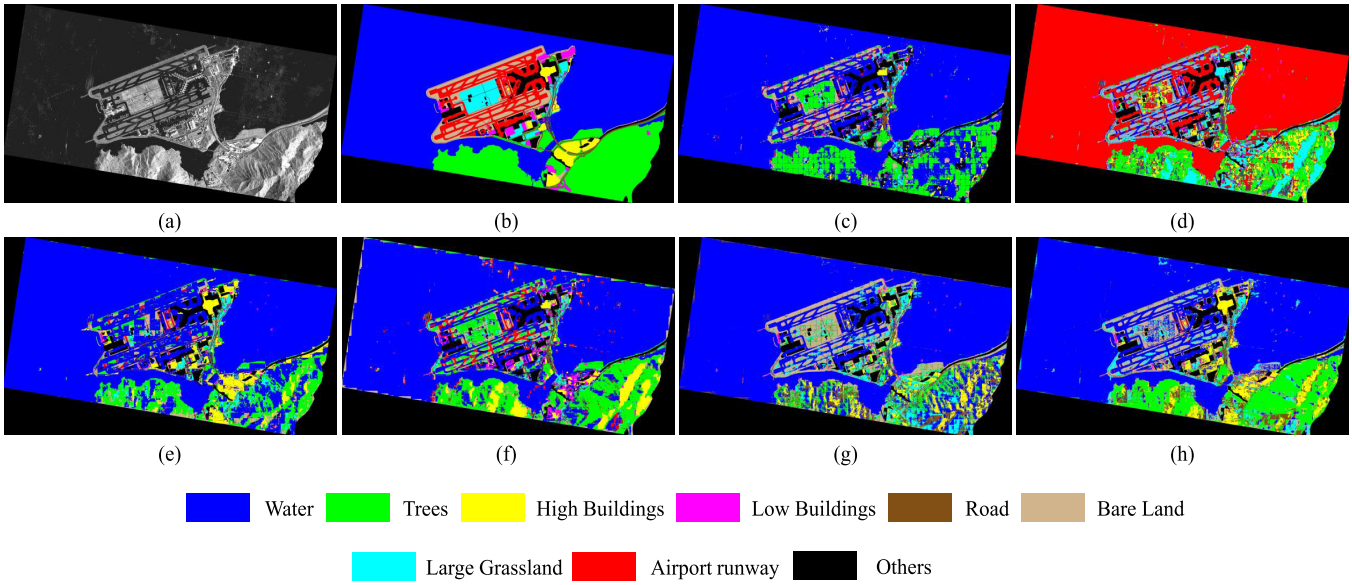



Fig. 13. Classification maps of different methods in Hong Kong Airport. (a) SAR image. (b) Ground Truth. (c) DeepLabv3+. (d) CCNR. (e) SegFormer. (f) MAE. (g) Baseline (DDPM). (h) CDM.

discontinuous and fragmented predictions. In contrast, the CDM method effectively alleviates the issue of discontinuous recognition and achieves the highest performance across nearly all evaluation metrics among the compared methods. This demonstrates that, even in the presence of a large number of land cover categories, the proposed method is still capable of effective learning, and the extracted features exhibit strong discriminative power. By comparing the ground truth with the SAR image, it becomes evident that the brightness and texture characteristics of the airport runway are highly similar to those of water. As a result, identifying the airport runway under limited sample conditions poses a significant challenge. Even with extensive pretraining on large-scale datasets, it remains difficult for models to effectively learn the representative features of the airport runway.

According to the experimental results in Table IX, the CDM excels across multiple evaluation metrics, significantly outperforming other comparative methods such as DeepLabv3+, CCNR, SegFormer, MAE, and Baseline-ddpm. Specifically, CDM achieved an OA of 0.776, a kappa coefficient of 0.597, and an FWIoU of 0.674, all ranking highest among the compared models. Although its MIoU is 0.321, slightly lower than CCNR's 0.361, it nevertheless maintains a high standard. Furthermore, CDM demonstrates balanced and exceptional performance in $F1$ scores across various categories, with an $F1$ score of 0.391 for the high buildings category, 0.639 for bare land grassland, and as high as 0.926 for water. In conclusion, the CDM surpasses other supervised and self-supervised methods in multicategory scenarios.

### D. Ablation Study

To further investigate the role of contrast strategies and CRFs in the diffusion model, this ection conducts ablation experiments on three scenes: PoDelta, Mississippi (north), and Pohang, within the CDM framework. These experiments aim to elucidate the impact and effectiveness of contrast strategies and CRFs in the diffusion model. In addition to providing the imaging parameters listed in Table I, the sensor also includes explicit system parameters such as imaging mode, pitch angle, radiation enhancement, and other implicit parameter information. The sensor therefore has a large influence on the pixel distribution of SAR images. The three SAR images used in the ablation study correspond to distinct sensors, allowing in

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TABLE IX
CLASSIFICATION RESULTS ON HONG KONG AIRPORT

| Method | F1 Score | | | | | | | | OA | kappa | MIoU | FWIoU |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Water | Trees | Road | High buildings | Low buildings | Bareland grassland | Large grassland | airport runway | | | | |
| Deeplabv3+ | 0.490 | 0.043 | **0.525** | 0.293 | **0.588** | 0.389 | 0.356 | 0.247 | 0.431 | 0.236 | 0.324 | 0.270 |
| CCNR | 0.782 | 0.393 | 0.614 | 0.305 | 0.219 | 0.618 | **0.488** | **0.65** | 0.592 | **0.361** | 0.504 | 0.429 |
| SegFormer | 0.797 | 0.365 | 0.616 | 0.188 | 0.323 | 0.332 | 0.396 | 0.005 | 0.501 | 0.260 | 0.398 | 0.342 |
| MAE | 0.685 | 0.380 | 0.558 | 0.181 | 0.382 | 0.531 | 0.452 | 0.405 | 0.515 | 0.298 | 0.414 | 0.350 |
| Baseline (DDPM) | 0.897 | 0.271 | 0.198 | 0.225 | 0.158 | 0.612 | 0.162 | 0.002 | 0.694 | 0.228 | 0.436 | 0.563 |
| CDM | **0.926** | **0.712** | 0.311 | **0.391** | 0.202 | **0.639** | 0.227 | 0.028 | **0.776** | 0.321 | **0.597** | **0.674** |

TABLE X
ABLATION STUDY OF DIFFERENT COMPONENTS

| DDPM | Contrast Learning | CRFs | PoDelta | | | | Mississippi (north) | | | | Pohang | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | OA | kappa | MIou | FWIou | OA | kappa | MIou | FWIou | OA | kappa | MIou | FWIou |
| ✓ | | | 0.914 | 0.813 | 0.587 | 0.882 | 0.804 | 0.522 | 0.537 | 0.706 | 0.630 | 0.498 | 0.382 | 0.490 |
| ✓ | ✓ | | 0.973 | 0.939 | 0.786 | 0.950 | 0.885 | 0.693 | 0.661 | 0.803 | 0.688 | 0.574 | 0.457 | 0.557 |
| ✓ | | ✓ | 0.957 | 0.906 | 0.725 | 0.932 | 0.894 | 0.687 | 0.707 | 0.807 | 0.777 | 0.683 | 0.514 | 0.647 |
| ✓ | ✓ | ✓ | **0.978** | **0.952** | **0.850** | **0.958** | **0.914** | **0.756** | **0.738** | **0.842** | **0.814** | **0.734** | **0.580** | **0.700** |

a comprehensive exploration of the modules' generalization capabilities across various scenes.

Table X shows the four overall indices for land cover classification under different components for the three scenes. Contrast learning involves integrating contrast learning concepts into the diffusion pretraining process to enforce the consistency of feature activation in noisy images across different steps. This approach also utilizes a training strategy that incorporates learnable variance and cos noise scheduling. Although self-supervised learning frameworks effectively extract high-level semantic features, the inherent complexity and noise characteristics of SAR imagery often result in inaccuracies in the initial classification results, particularly in the details along land cover boundaries. CRFs address these shortcomings by postprocessing the classification probability maps to refine the classification results, effectively mitigating the limitations of the front-end model and enhancing overall classification accuracy. In this article, the CRFs module was used to remove isolated clutter from the prediction result and smooth the classification results to improve classification performance. The inclusion of either contrast learning or the CRFs module results in a performance improvement of more than 4% across all four overall indices compared to DDPM. In most scenes, the use of CRFs can provide a greater boost than contrast learning. This is due to the introduction of noisy information into the single-step denoising diffusion process, where the image features contain noise leading to clutter in the prediction maps. Unlike CRFs, which rely on context to enforce consistency between pixels, this approach eliminates clutter to achieve significant improvement. Furthermore, the combined use of contrast learning and CRFs modules further improves classification accuracy. This highlights the essential role of integrating contrast learning and CRFs into the diffusion model for land cover classification, ensuring the provision of discriminative feature representations tailored to downstream tasks. CRFs offer a distinct optimization mechanism by modeling higher order dependencies among pixels. Unlike other components, CRFs effectively balance global context and local details, enhancing spatial coherence in classification results while refining boundary precision. Given that no other module within the current framework can replicate these capabilities, the integration of CRFs is essential and irreplaceable for achieving superior classification performance.

### *E. Classification Results Under Different Training Quantity*

We conducted experiments on the Rosenheim dataset with varying training data sizes (4, 8, 12, 16, 10, 24, and 28 samples) to evaluate the stability of the CDM model. Performance was rigorously quantified using established metrics: kappa, MIOU, OA, and FWIOU. The results in Fig. 14 unequivocally demonstrate that the CDM model achieves superior scores across all metrics even under extreme low-data regimes (4 samples), highlighting its exceptional few-shot learning capability. However, increasing the amount of data beyond 16 samples yielded diminishing returns, with statistically insignificant improvements in all evaluation criteria. This saturation effect conclusively justifies our selection of 16 samples as the optimal trade-off between annotation cost and model performance in the final implementation.

### *F. Discussion*

Based on the above comparison of land cover classification experiments on six SAR scenes, we systematically and comprehensively analyzed the experimental results.

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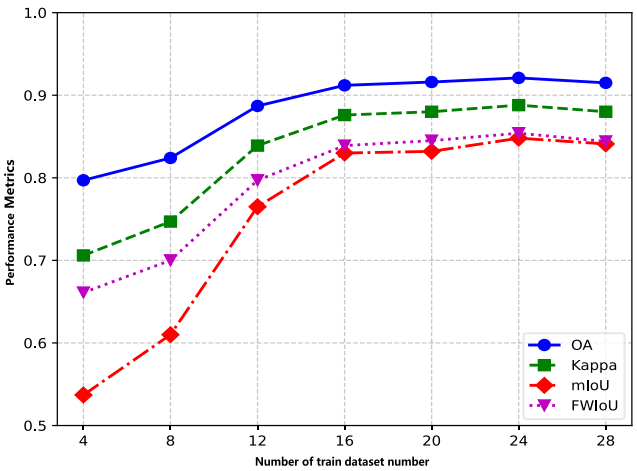



Fig. 14. Index chart with data volume under CDM method.

DeepLabv3+ and CCNR exhibit significant overfitting issues caused by supervised training with a limited number of pixel-level labeled samples, resulting in numerous blocking artifacts. SegFormer mitigates blocking artifacts and overfitting issues through contextual learning facilitated by a global self-attention mechanism. However, it is susceptible to clutter caused by speckle in SAR images. Compared to SegFormer, MAE demonstrates improved capability in learning general terrain features through extensive pretraining and shows less susceptibility to speckle interference. However, MAE's reliance on large-scale mask repair as a pretext task weakens its ability to perceive terrain boundaries. CDM employs a contrast-diffusion pretraining process, enabling comprehensive global data distribution learning compared to MAE. This approach significantly enhances the perception of terrain boundaries. However, the supervised objective of pretraining does not align with the downstream land cover classification task, necessitating the introduction of noisy information into CDM. Consequently, CRFs are employed to address this challenge. In conjunction with the analyses from the ablation study, we posit that global data distribution learning and CRF smoothing processing are pivotal in achieving outstanding land cover classification performance in CDM.

## V. Conclusion

In this article, a novel generative self-supervised learning land cover classification framework called CDM using a contrast-diffusion architecture is proposed to mine feature representations of SAR images. The CDM harnesses potential and learns intricate data distributions from extensive unlabeled datasets, aiming to extract common feature representations, diminish label dependency, and enhance generalization capabilities. To improve the CDM, we propose DRRE module and contrast learning strategy, which enforce the consistency of feature activation and facilitate targeted feature interfaces for classification tasks. Specifically, the feature activation of the CDM output layer is enhanced through DRRE, followed by the application of contrastive learning to promote convergence of features across different diffusion steps. The effectiveness of the proposed method is validated on the task of land cover classification of 6 SAR scenes under the pretraining of 18 broad high-resolution SAR scenes.

In the future, we aim to analyze and deconstruct the feature activation of the diffusion model at different steps in more depth to obtain a more general and essential feature representation of SAR images. We hope that our work will inspire more remote sensing researchers to introduce denoising diffusion models into the SAR field.

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